



THE COPPERBELT UNIVERSITY
SCHOOL OF GRADUATE STUDIES

CS 594: CLIMATE MODELLING AND DATA ANALYSIS

PRACTICAL 03:

**SPATIAL CLIMATE TREND ANALYSIS OVER ZAMBIA USING CMIP6 HISTORICAL
DATA (1981–2010)**

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1. Introduction

Climate change is associated with long-term changes in climatic variables such as temperature and precipitation. Examining these variables over time is important for identifying whether particular locations are experiencing warming, cooling, increased rainfall, or decreased rainfall. Spatial climate trend analysis is especially useful because climate changes are not necessarily uniform across a country. Climate models provide gridded climate data that can be used to navigate historical climate variability and trends. However, their spatial resolution may be coarse compared with administrative boundaries, making spatial preprocessing and careful interpretation necessary.

This study analyses historical CMIP6 climate data from the MIROC6 model over Zambia from 1981–2010. Three daily climate variables were analysed: maximum near-surface air temperature (*tasmax*), minimum near-surface air temperature (*tasmin*), and precipitation (*pr*). The analysis aimed to determine the magnitude, direction, and statistical significance of spatial climate trends across Zambia. Sen's slope estimator quantified the magnitude of change per decade, while the Mann–Kendall test determined whether the detected trends were statistically significant.

2. Methods

2.1. Climate and Spatial Data

Daily historical climate data were obtained from the CMIP6 MIROC6 climate model for the period 1981–2010. The analysis considered daily maximum temperature (*tasmax*), daily minimum temperature (*tasmin*), and precipitation (*pr*). The climate datasets were provided as NetCDF files and processed in R using the terra package. Zambia Level-2 administrative boundaries were obtained from the GADM spatial dataset and loaded from the *gadm41_ZMB_2.shp* shapefile. The boundary dataset contained district-level polygons used to define the spatial study area.

2.2. Unit Standardisation

Temperature data were originally expressed in Kelvin and were converted to degrees Celsius to improve interpretation:

$$T(^{\circ}C) = T(K) - 273.15$$

Precipitation was provided as a flux in $\text{kg m}^{-2} \text{ s}^{-1}$ and converted to mm/day:

$$P_{\text{mm/day}} = P_{\text{kg m}^{-2} \text{ s}^{-1}} \times 86,400$$

The conversion to mm/day was necessary because precipitation can then be meaningfully accumulated into annual rainfall totals.

2.3. Spatial Processing: Cropping and Masking

The climate rasters and Zambia administrative boundaries were checked for CRS compatibility. The rasters were then cropped to the Zambia extent and masked to exclude cells outside the study area.

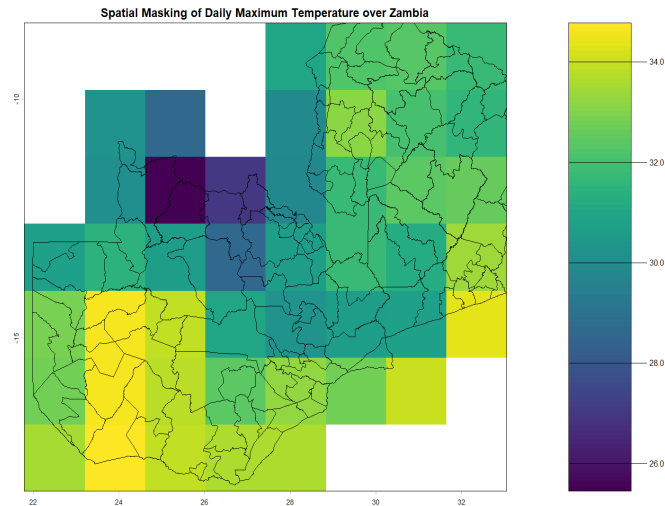


Figure 2.1. Spatial masking of CMIP6 daily maximum temperature data to the Zambia administrative boundary.

Daily temperature observations were aggregated into annual mean Tmax and Tmin, while daily precipitation was summed to produce annual precipitation totals. Annual values were retained only when at least 300 valid daily observations were available, reducing the influence of incomplete records.

2.4. Trend Analysis

Climate trends were calculated for each valid grid cell using Sen's slope, while the Mann–Kendall test was used to determine statistical significance. Sen's slopes were converted to decadal rates, giving temperature trends in °C/decade and precipitation trends in mm/decade. A trend was classified as statistically significant when $p < 0.05$.

3. Results

The spatial trend analysis showed clear differences between temperature and precipitation trends across Zambia during 1981–2010. Both maximum and minimum temperatures showed positive trends across all analysed grid cells, while precipitation exhibited both increasing and decreasing trends. The Sen's slope ranges from the analysis were 0.103–0.966 °C/decade for *tasmax*, 0.187–0.456 °C/decade for *tasmin*, and –116.263–67.736 mm/decade for precipitation.

3.1. Maximum Temperature Trends

Maximum temperature (*tasmax*) showed an increasing trend across the study area. Sen's slope ranged from 0.103 to 0.966 °C/decade, with a mean of 0.457 °C/decade and a median of 0.472

°C/decade. Approximately 55.56% of the analysed grid cells had statistically significant trends ($p < 0.05$). These results indicate an overall warming trend in maximum temperature across Zambia, although the magnitude and statistical significance of warming varied spatially.

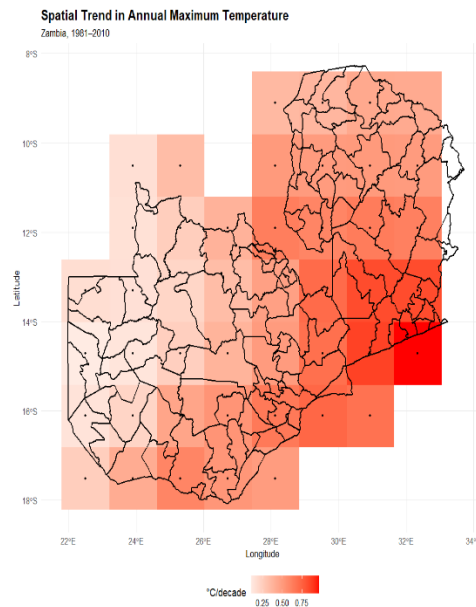


Figure 3.1. Spatial trend in annual maximum temperature (°C/decade) over Zambia, 1981–2010.

3.2. Minimum Temperature Trends

Minimum temperature (*tasmin*) also showed a positive trend across all analysed grid cells. Sen's slope ranged from 0.187 to 0.456 °C/decade, with a mean of 0.301 °C/decade and a median of 0.288 °C/decade. Approximately 95.56% of grid cells showed statistically significant trends. The high percentage of significant cells indicates that the increase in minimum temperature was particularly widespread and consistent across the study area.

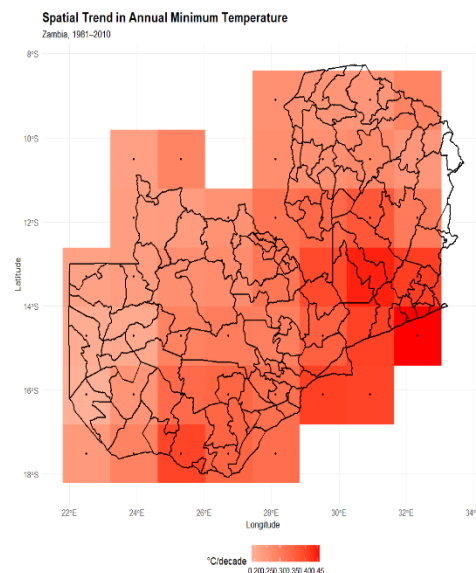


Figure 3.2. Spatial trend in annual minimum temperature ($^{\circ}\text{C}/\text{decade}$) over Zambia, 1981–2010.

3.3. Precipitation Trends

Unlike temperature, precipitation showed both positive and negative trends across Zambia. Sen's slope ranged from -116.263 to 67.736 mm/decade, with a mean of -23.457 mm/decade and a median of -34.151 mm/decade. The negative mean and median indicate that the analysed grid cells generally tended to decrease in annual precipitation.

However, only 4.44% of grid cells showed statistically significant precipitation trends ($p < 0.05$). Therefore, although the spatial results indicate an overall tendency towards decreasing precipitation, the trend was not statistically significant across most of the study area.

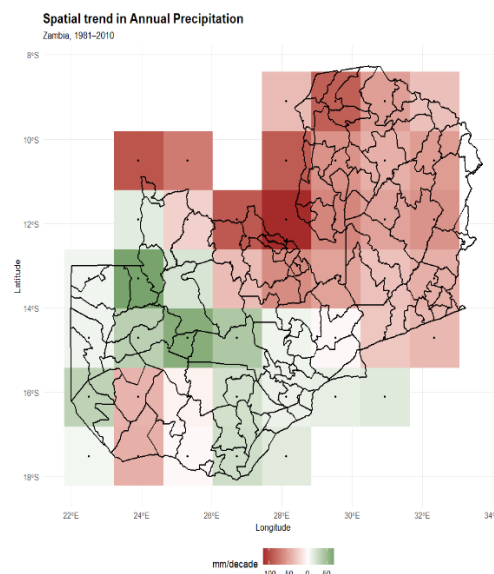


Figure 3.3. Spatial trend in annual precipitation (mm/decade) over Zambia, 1981–2010.

3.4. Summary of Spatial Trends

The results indicate a clear warming pattern across Zambia, with both maximum and minimum temperatures showing positive trends. Minimum temperature had the most widespread significant change, with 95.56% of grid cells, versus 55.56% for maximum temperature. Precipitation showed a negative trend of -23.457 mm/decade, implying decreasing rainfall, but only 4.44% of the grid cells were significant, suggesting inconsistent rainfall changes.

Table 3. Climate Modelling Practical 1. Summary statistics of Sen's slope and percentage of statistically significant grid cells for tasmax, tasmin and precipitation over Zambia (1981–2010).

Variable	Minimum slope	Maximum slope	Mean slope	Median slope	Significant cells (%)
Tasmax ($^{\circ}\text{C}/\text{decade}$)	0.103	0.966	0.457	0.472	55.56

tasmin (°C/decade)	0.187	0.456	0.301	0.288	95.56
Precipitation (mm/decade)	-116.263	67.736	-23.457	-34.151	4.44

4. Discussion

The results indicate a clear warming pattern across Zambia between 1981 and 2010. Both maximum and minimum temperatures increased, although minimum temperature showed the most widespread statistically significant change. With 95.56% of tasmin grid cells showing significant trends, compared with 55.56% for tasmax, the results suggest that minimum temperatures warmed more consistently across the study area. Precipitation showed a different pattern, with both increasing and decreasing trends across Zambia. Although the negative mean slope suggests an overall tendency towards declining annual precipitation, only 4.44% of grid cells showed statistically significant trends. Therefore, interpret the precipitation results cautiously, as most observed changes are not statistically significant based on the Mann–Kendall test.

The spatial patterns also demonstrate the importance of analysing climate change geographically rather than relying only on national averages. However, the relatively coarse resolution of the MIROC6 data limits the ability to represent detailed climate variations at district level.

5. Conclusion

The study identified a clear warming trend across Zambia from 1981–2010, with particularly widespread and statistically significant increases in minimum temperature. Maximum temperature also increased, while precipitation generally showed a declining tendency but with limited statistical significance. Overall, the findings demonstrate that temperature changes were more consistent than precipitation changes across Zambia during the study period.